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25 April 1980
SUPERSEDING
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28 September 1978

MILITARY SPECIFICATION

SURVIVAL KIT CONTAINER, AIRCRAFT SEAT, WITH OXYGEN; GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for an aircraft rigid seat survival kit (RSSK) containing emergency oxygen for use by aircrewmen during normal flight operations and during and following emergency egress. The kit also contains survival equipment.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specififcation to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-383	-Plastic Material, Polyester Resin, Glass Fiber Base, Low Pressure Laminated
UU-T-81	-Tag, Shipping and Stock
PPP-B-636	-Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, ESSD, Code 93, Lakehurst, New Jersey 08733 by using the self-addressed Standard Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1660

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MIL-P-116	-Preservation, Packing, Methods of
MIL-R-5001	-Rubber, Cellular Sheet, Molded and Hand Built Shapes, Latex Foam
MIL-E-5272	-Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-P-7105	-Pipe Threads, Taper, Aeronautical National Form, Symbol, Anpt, General Requirements for
MIL-C-7219	-Cloth, Duck, Nylon
MIL-G-7601	-Gage, Emergency Oxygen, Type L-2
MIL-S-7742	-Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-C-7905	-Cylinders, Compressed Gas, Nonshatterable
MIL-V-7908	-Valve, Check, Aircraft Low Pressure Oxygen
MIL-A-8625	-Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-V-8965	-Valve Core, Oxygen, High Pressure
MIL-V-9050	-Valve, Oxygen, Cylinder, High Pressure
MIL-P-9400	-Plastic Laminate and Sandwich Construction Parts, Aircraft Structural, Process Specification Requirements
MIL-R-17852	-Reducer, Oxygen Pressure (On-off Type)
MIL-H-19089	-Harness, Integrated Parachute

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STANDARDS

FEDERAL

FED-STD-406	-Plastic, Methods of Testing
FED-STD-595	-Colors

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-Marking for Shipment and Storage
MIL-STD-130	-Identification Marking of U.S. Military Property
MIL-STD-143	-Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-794	-Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-889	-Dissimilar Metals
MS14174	-Dropline, Rigid Seat Survival Kit
MS14175	-Dropline/Boot Assembly, Rigid Seat Survival Kit

AIR FORCE - NAVY AERONAUTICAL

AN833	-Elbow, Flared Tube and Bulkhead 90 Degrees

DRAWINGS

NAVAL AIR DEVELOPMENT CENTER

CL204D2	-Bracket Assembly, AN/URT 33 Radio Beacon
C1223A3	-Elbow, Flared Tube, 5/16 to 1/4 Tubing, RSSK-8D

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PUBLICATIONS

NAVAL AIR ENGINEERING CENTER

ACEL Report No. 533 -Anthropometry of Naval Aviators 1964

NAVAL AIR SYSTEMS COMMAND

Aviation Crew Systems Manual

NAVAIR 13-1-6.3 -Survival Kits and Items

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The rigid seat survival kits furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3).

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4.).

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein or in the applicable detail specifications shall be selected in accordance with MIL-STD-143.

3.4 Materials and components. Materials and components shall conform to applicable specifications and as specified herein and in the applicable detail specifications. Materials which are not covered by applicable specifications or which are not specifically described herein, or in the applicable detail specifications, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.4.1 Metal parts. All metal parts shall be of a corrosion resistant material or treated in a manner to render them adequately resistant to corrosion.

3.4.1.1 Dissimilar metals. Dissimilar metals, such as defined by MIL-STD-889 shall not be used in intimate contact with each other unless suitably protected against electrolytic corrosion with protective coatings.

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3.4.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all components except where magnetic materials offer definite advantages.

3.4.3 Nonmetallic materials.

3.4.3.1 Elastomer components. The elastomer components shall be composed of an ozone resistant composition.

3.4.3.1.1 Age. Elastomer components shall be not more than 12 months old from the cure date of manufacture to the date of delivery to any Government Service or to any airframe or accessory manufacturer.

3.4.3.2 Fiberglass. Fiberglass, if used in the construction of the container, shall be manufactured in accordance with the requirements of MIL-P-9400. The fiberglass material shall conform to the requirements of L-P-383. The test slabs of the basic container when tested as specified in 4.8.13 shall exhibit the properties of resin content. The maximum resin content shall be 33 percent.

3.4.4 Protective treatment. When materials are used in the construction of the rigid seat survival kit that are subject to corrosion in oxygen, salt air, or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that shall in no way prevent compliance with the performance requirements of this specification. Any protective coating or finish that shall chip or scale during normal service life, or under extremes of atmospheric conditions, shall not be used.

3.4.4.1 Aluminum alloy parts. Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625.

3.4.5 Fungus-proof materials. Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.6 Standard parts. MS and AN standard parts shall be used where they suit the purpose, and shall be identified by their part number. In applications for which no suitable standard part is in effect on the date of invitation to bid, commercial parts may be used, provided they conform to the requirements of this specification.

3.4.7 Deleterious fumes. Materials or material treatments which shall cause toxic reaction in this application or transfer odor of an obnoxious or unpleasant type shall not be used.

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3.4.8 Threads.

3.4.8.1 Screw threads. Screw threads shall conform to MIL-S-7742.

3.4.8.2 Pipe threads. Pipe threads shall conform to MIL-P-7105.

3.5 Environmental conditions. Unless otherwise specified in the applicable detail specification, the rigid seat survival kit shall function as designed when exposed to any likely combination of the following environmental conditions:

a. Temperature from -65 to +160° F.

b. Humidity as encountered in all-weather operation.

c. Salt spray as encountered in salt water operations.

d. Sand and dust as encountered in desert operations.

e. Vibration within the limits of tests, as described in 4.8.12.1.

3.6 Design and construction. The RSSK shall be designed and oriented in such a manner as to accommodate and provide a maximum of comfort, convenience, and freedom from hazard for the aircrewman (lower limit 3 percentile man, upper limit 98 percentile man, as specified in ACEL Report No. 533). The seat kit shall be designed or oriented in such a manner as to integrate satisfactorily with the airframe, seat, and aircrewman's personnal equipment. The seat kit shall be designed to accommodate an emergency bailout oxygen system and applicable survival equipment within the configuration envelope. The seat kit shall be designed such that any hardware, handles, or other equipment protruding above the seat kit shall not cause pressure points or discomfort during service usage, ejection, or possible crash loads. The container shall provide attachment to the aircrewman's torso harness such that it will stay with the aircrewman during ejection and fall. The seat kit when installed in the seat shall be compatible with the in-service adjustment of the seat and maintain the in-service pilot's eye reference level requirement and accommodate the 3 to 98 percentile aircrewman. As a design goal, in aircraft where the in-service adjustment of the ejection seat is physically limited by the anthropometry of the 3 or 98 percentile aercrewman, the design of the seat kit shall compensate, where possible, the adjustment limitations on the ejection seat to allow for full-up and full-down adjustment of the seat to the eye reference level. Such a design would require the approval of the Naval Air Systems Command. The seat kit shall be equipped with an aircraft-to-kit oxygen and

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electrical disconnect and the necessary tubing, fittings, and wiring for connecting the aircraft and the kit oxygen and electrical systems. The optimum design of the RSSK shall be such that the components affecting the ejection seat to RSSK and torso harness restraint subassembly shall conform to the general arrangement shown in figure 1.

3.6.1 Operation. The rigid seat survival kit shall be designed to allow the following operations:

a. Removal of the kit from the aircraft for servicing, packing, etc., without the use of special tools for detaching or attaching service lines from the aircraft to the kit and without repacking the container.

b. Normal abandonment of the aircraft after a flight, leaving the RSSK in the aircraft by actuating the quick release fittings of the restraint subassembly from the torso harness and the aircraft services quick disconnect oxygen hoses between aircrewman and the RSSK.

c. Ejection with automatic activation of the emergency bailout oxygen system and subsequent manual release of the release handle causing the bottom container to separate and fall away from the contoured container lid secured to the aircrewman. The bottom container as it falls actuates the inflation system of the life raft causing it to inflate. The raft and survival equipment shall be secured to the aircrewman by means of the dropline which is connected to the container bottom and to the container lid. This operation shall be accomplished in the following steps:

1. Firing of the ejection seat shall automatically activate the emergency bailout oxygen system in a manner as specified in 3.6.2.6.1.5.2. A manual means of actuation for the emergency oxygen system shall be provided in case the automatic fails and also to provide means of actuation should the ship's oxygen supply fail.

2. An upward pull on the release handle shall separate the container bottom from the container lid and aircrewman, and as the container bottom falls to full line stretch it shall actuate the life raft inflation system.

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The oxygen system which is in the container lid shall be connected and retained to the aircrewman.

3.6.2 Components. Unless otherwise specified, the rigid seat survival kit shall consist of the following components:

a. Basic container.

b. Comfort cushion.

c. Ejection seat to RSSK and torso harness restraint subassembly.

d. Release handle and container locking mechanism.

e. Container dropline.

f. Emergency bailout oxygen system.

g. Radio beacon bracket.

3.6.2.1 Basic container. The seat kit shall be designed to provide basically a cushioned seat conforming to the dimensions and general design as specified in the applicable detail specification. It shall be designed to permit the incorporation of an emergency oxygen bailout system and the container shall be designed to provide a maximum of interior space consistent with the limitations and requirements as specified herein and the physical limitations of the applicable ejection seat bucket. The basic container shall be composed of two parts consisting of the contoured container lid and a container bottom with means for locking and releasing the lid as specified in the detailed specification. The alignment of the fitting and mating surfaces shall be accurate to a degree that shall permit the proper function of the unit within the environmental conditions expected during the service life of the unit. The fit of the assembled components shall be such that upon actuation of the release handle, the container bottom shall separate and fall away easily from the contoured container lid.

3.6.2.1.1 Container bottom. The container bottom shall be designed to accommodate the life raft and applicable survival equipment within the maximum interior space consistent with configuration, strength, and other applicable requirements. The container bottom shall provide lid supports and shall be designed for attachment thereto of a release handle, locking mechanism, actuating cables and radio beacon bracket. The interior of the container bottom shall be free from any defect, burr or sharp edge, and the attachments for hardware components shall not damage or abrade the life raft or survival equipment under all conditions of operation, including vibration and performance tests.

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3.6.2.1.1.1 Radio beacon bracket. A radio beacon bracket manufactured in accordance with drawing CL204D2 shall be installed as specified in the detail specification.

3.6.2.1.2 Contoured container lid. The contoured container lid of the seat kit shall conform to the applicable detail specification. The lid shall be recessed to permit the storage of the emergency bailout oxygen system. The lid shall contain hardware for the retention of the emergency bailout oxygen system and shall hold the oxygen system secure during service use and in all performance tests specified herein.

3.6.2.1.3 Continer locking mechanism. The container locking mechanism specified in the detail specification shall be incorporated for attaching together the upper and lower parts of the container. The mechanism shall not require special tools to open or close the kit and shall separate the container bottom from the lid when the release handle is actuated as specified in 3.6.2.4. The mechanism shall not permit inadvertent separation or separation of the two parts of the kit container when the container is subjected to the conditions specified in 3.6.3.3, 4.8.3, 4.8.5.2, 4.8.6 and 4.8.7. The container locking mechanism, including all actuating cables, shall be properly guarded to prevent inadvertent actuation or damage by or to the container contents. The container bottom shall incorporate inspection slots for visually determining locking of the lid to the bottom. The inspection slots shall be covered with suitable clear mylar tape or equivalent.

3.6.2.2 Comfort cushion. The comfort cushion shall conform to the configuration of the container lid. The comfort cushion shall be attached to the container lid and shall contain provisions for being retained to the kit under all normal and emergency operations. The comfort cushion shall not cause interference with the aircraft control stick movement, or any operation or function of the survival kit or ejection seat.

3.6.2.2.1 Comfort cushion outer covering. The comfort cushion outer covering material (fabric) shall conform to MIL-C-7219, type III. The color shall approximately match USAF sage green number 1535.

3.6.2.2.2 Comfort cushion material. The cushioning material shall allow for bottoming of the aircrewman during ejection without causing amplification of the ejection loads. The cushion material shall be one inch thick conforming to MIL-R-5001, type II, class firm, grade A.

3.6.2.3 Ejection seat to RSSK and torso harness subassembly. The RSSK shall be designed to include a torso restraint system which shall be compatible with MIL-H-19089 integrated torso suit harness, required personnel environmental protective and flight garments and survival equipment. The RSSK shall provide a means for securing the container to the aircrewmwn and if applicable, the container to the seat, until separation

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or release functions are performed. The lap belt portion shall consist of webbing, non-slip adjustment hardware and other components as specified in figure 1. The restraint subassembly shall provide adjustment to accommodate 3 to 98 percentile aircrewman without inducing pressure points or causing interference with escape and survival functions.

3.6.2.4 Release handle and container locking mechanism. The basic container shall contain means for manually releasing and separating the bottom from the container lid. The release handle shall be configured as specified in the detail specification and shall be located on the container bottom on the right side. The handle shall not extend past the front of the container. It shall be in an easily accessible position to allow for actuation by the 3 to 98 percentile man while the man is suspended in the parachute harness following ejection. The location and design shall not cause uncomfortable pressure points. Unless otherwise specified, the release handle and container locking mechanism shall be such that the container shall unlock only when an upward pull is exerted on the handle. The design shall be such that a downward force shall not unlock the container (see 4.8.2). The container locking mechanism shall be so designed that the first half inch of travel, upon pulling the release handle, shall unlock the two container halves. A continued pull shall cause the handle to strip free from the container. The force required to actuate the handle and unlock the lid from the container bottom shall be 20 ± 10 pounds applied to the handle perpendicular to the base of the handle at a point between 4 and 4-1/4 inches forward of the fulcrum point of the handle. The force required to separate the handle from the container locking mechanism shall be a maximum of 30 pounds. Only a moderate steadily applied one-handed force shall be required to reinstall the handle and secure the kit during maintenance. The mechanism shall be designed to prevent insertion of the handle without locking the container.

3.6.2.5 Container dropline. The seat kit shall incorporate a dropline conforming to MS14174. Provisions shall be made for the attachment of the dropline between the container lid and the container bottom. Upon actuation of the release handle and opening of the kit, the dropline shall allow the container bottom to drop 25 feet, suspending the container bottom, the inflated life raft and survival equipment package from the container lid or harness which remains attached to the aircrewman. The dropline assembly incorporates provisions for attaching and inflating the one-man life raft. The provisions for life raft inflation shall be such that as the container bottom falls away from the lid and at full extension of the dropline, the attachment between the dropline and the life raft inflation system shall actuate the inflation system which shall inflate the life raft. The attachment shall be such that there shall not be any possibility of inadvertent inflation of the life raft due to vibration or movement during normal service use. The dropline shall be reeved as flat as possible into two boots as detailed in MS14175. The dropline shall not interfere with inflation or deployment of the raft from the container regardless of the method of folding the raft.

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3.6.2.6 Emergency bailout oxygen system. The rigid seat survival kit emergency bailout oxygen system shall be designed to be installed and retained in the seat kit contoured container lid. The oxygen system shall be firmly secured in the container lid in such a manner that it shall not work loose or cause leakage or damage to the system when the container is subjected to the performance requirements and tests or during normal service use. The emergency bailout oxygen system shall remain with the aircrewman during his descent following ejection in that it shall be secured to the container lid which is attached to the aircrewman. When actuated either manually or automatically upon ejection, the reducer shall regulate the emergency oxygen pressure. A one-way check valve shall be provided to prevent the emergency oxygen from flowing in the direction of the aircraft's oxygen supply. Electrical communications are also tied into the seat kit through the applicable disconnect.

3.6.2.6.1 Components. The emergency bailout oxygen system shall consist of the following components:

a. Emergency oxygen cylinder.

b. Oxygen pressure reducer (on-off type).

c. Filler valve.

d. Pressure gage.

e. Oxygen manifold assy.

f. Check valve.

g. Relief valve.

h. Integrated oxygen/communications hose assembly.

i. Actuation mechanism.

3.6.2.6.1.1 Emergency oxygen cylinder. The emergency oxygen cylinder shall be capable of containing oxygen at a pressure of 1800 psi. The cylinder shall be in accordance with MIL-C-7905 and shall be mounted in the contoured container lid. The cylinder shall be painted green approximately matching color number 14187 of FED-STD-595, and shall have the following marking on it in white approximately matching color number 17875 of FED-STD-595: "Breathing oxygen, 1800 psi (12.4MPa)". The cylinder shall supply oxygen for respiration in the event of bailout or failure of aircraft's supply oxygen.

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3.6.2.6.1.2 Oxygen reducer. The oxygen reducer in the emergency bailout oxygen system shall conform to the performance requirements only of MIL-R-17852, type I, except that the manual emergency oxygen actuator shall be attached to an O-ring type handle located as shown in the detail specification.

3.6.2.6.1.2.1 Filler valve. The emergency oxygen bailout system shall contain a filler check valve conforming to MIL-V-8965 for oxygen refilling purposes. The design, location and use of the filler check valve shall not require the use of special tools. The valve shall be readily accessible for refilling through an access hole in the contoured container lid. The access hole shall be 1.25 inches in diameter and located in the lid over the filler valve. The access hole shall be capped with a retainer cap.

3.6.2.6.1.2.2 Pressure gage. The emergency bailout oxygen system shall contain a pressure gage conforming to MIL-G-7601 for indicating oxygen pressure. The gage shall be visible for preflight check and shall be located beneath the inspection cutout in the contoured container lid.

3.6.2.6.1.3 Oxygen manifold assembly. The emergency bailout oxygen system shall contain a relief valve and a check valve in the oxygen manifold assembly.

3.6.2.6.1.3.1 Check valve. The check valve of the oxygen manifold assembly shall conform for performance requirements only to MIL-V-7908.

3.6.2.6.1.3.2 Relief valve. The relief valve of the oxygen manifold assembly shall conform for performance requirements only to MIL-V-9050, type 5, except for the following:

Tests	Modifications
Low temperature	Three hours at -65° F.
High Temperature	Three hours at $+160^{\circ}$ F.
Sustained Flow	30 liters per minute (lpm) for 15 minutes.
Vibration	Six hours, double amplitude 0.018, frequency 300 to 3,000 cycles per minute (cpm).

3.6.2.6.1.4 Oxygen hose with quick disconnect. The rigid seat survival kit shall contain the oxygen hose assembly with quick disconnect conforming to the applicable detail specification.

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3.6.2.6.1.5 Actuation mechanism. The oxygen system shall be designed to use the emergency supply/actuator assembly and shall provide manual and automatic actuation.

3.6.2.6.1.5.1 Manual actuation. Manual actuation shall be accomplished by a pull on a ring located on the left thigh extension. The ring shall be attached by cable to the release valve actuator and shall be actuated by a 20 ± 10 pound force. Color of the ring shall be green.

3.6.2.6.1.5.2 Automatic actuation. Automatic actuation shall occur simultaneously with seat ejection. Automatic actuation shall be accomplished by a cable which is attached to the airframe and by means of a break-away disconnect which in turn is connected to another cable attached to the on-off actuator of the emergency oxygen pressure reducer, and shall be actuated by a 20 ± 10 pound force acting on the cable.

3.6.2.6.1.6 Electrical communications system. The electrical commumications system of the rigid seat survival kit shall conform to the applicable detail specification. The system shall be compatible and integrate with the aircraft system and shall be connected to and complete the system through the commumication leads of the oxygen hoses, coupled at the quick disconnects.

3.6.3 Design load requirements. The rigid seat survival kit including the equipment shall be capable of sustaining the following loads:

3.6.3.1 Static loads. The rigid seat survival kit containing the emergency bailout oxygen system, life raft, and survival equipment package, closed and locked, shall sustain the specified loads without any evidence of material failure or malfunction of the container or installed equipment, except as indicated.

3.6.3.1.1 Basic container. The basic container of the rigid seat survival kit shall sustain the following uniform loads for a period of 2 seconds when tested as specified in Section 4:

a. Vertical compressive load - 7250 pounds proof

b. Fore and aft compressive load - 2000 pounds proof
3000 pounds ultimate

3.6.3.1.2 Container separation. The container lid shall not separate from the container bottom when subjected to the loads specified in the applicable detail specification.

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3.6.3.1.3 Ejection seat to RSSK and torso harness restraint subassembly. The restraint subassembly shall sustain the loads without failure as specified in 3.6.3.2, 3.6.3.3 and 4.8.5.

3.6.3.2 Crash loads. The rigid seat survival kit shall sustain the crash loads specified in 4.8.7 with an acceptable amount of structural container collapse. The acceptable level of structural container collapse shall be as determined by the testing facility with consideration given to the possible injurious effects the structural collapse could cause to the aircrewman and damage to the container contents. The ejection seat to RSSK and torso harness subassembly shall not fail and the container shall separate upon actuation of the release handle and shall conform to the requirements of 4.8.2, 4.8.4.1 and 4.8.8.1.

3.6.3.3 Windblast loads. The rigid seat survival kit shall sustain the windblast loads specified in the detail specification.

3.6.4 Compatibility requirement. The rigid seat survival kit shall be compatible with the ejection seat for which it is designed and shall maintain the in-service adjustment of the seat and maintain the in-service pilot's eye reference level accommodating the 3 to 98 percentile aircrewman.

3.7 Performance. The rigid seat survival kit shall meet all the performance requirements of this specification when tested as specified in Section 4 and applicable detail specification.

3.8 Size. The rigid seat survival kit shall conform to the dimensions specified in the applicable detail specification.

3.9 Weight. The weight of the rigid seat survival kit shall be as specified in the applicable detail specification.

3.10 Interchangeability. All parts of the rigid seat survival kit having the same manufacturer's part number shall be completely interchangeable with respect to installation and performance. Seat kits manufactured with the same manufacturer's part number shall have interchangeability between container bottoms, contoured lids and comfort cushions.

3.11 Maintenance. The design of the rigid seat survival kit shall be such as to be easily disassembled and reassembled for service and maintenance of the components without the use of special tools.

3.12 Color. The basic container color shall be green approximately matching color number 34128 of FED-STD-595. Color of anodizing shall approximate the aforesaid green on all aluminum components on the outside of the container, except tubing and the operating hardware which may be natural. Slight color deviation from the aforesaid green on fabric components is permissible between mill runs.

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3.13 Identification and marking. Identification markings shall conform to MIL-STD-130. Identification plates shall be securely attached and shall contain the following information:

Rigid Seat Survival Kit (dash type)
Specification
Manufacturer's part number
Manufacturer's serial number
Date of manufacture
National stock number
Contract number
Serial number (see 6.2)

3.14 Warning tag. A red warning tag conforming to UU-T-81, type B, not smaller than size 4 and strung with twine, shall be securely tied to restraint subassembly of the RSSK. The tag shall contain the following information:

WARNING

The rigid seat survival kit shall not be used until subjected to the inspection requirements of NAVAIR 13-1-6.3, Aviation Crew Systems Manual.

3.15 Workmanship. Fabrication shall be in accordance with the highest standards of commercial and aircraft practices. The non-fibrous components shall not be misaligned or distorted nor contain any corrosion, scale, pit, dent, crack, nick, burr, sliver, or sharp or rough edge. The fibrous components shall not contain any tear, cut, mend, needle chew, burn, spot, stain, nor any portion stiffened or scorched by any process of manufacture. The kit shall not contain any non-specified hole and drawn or molded parts shall be of equal thickness throughout, except where increase strength is required; however, laminated glass fiber reinforced material shall have no excess resin buildup or inconsistency of glass fibers in corner areas. Particular attention shall be given to alignment of parts and tightness of assemblies. Because of the emergency and life support use of these kits, the importance of providing a product of uniformly excellent quality cannot be overemphasized. The kits shall be uniform in quality and shall be free from irregularities or defects which could adversely affect performance, reliability, or durability. The kits shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of the rigid seat survival kit shall be classified as follows:

a. Qualification inspection. Qualification inspection consists of examinations and tests performed on rigid seat survival kits which are submitted for approval as qualified products (see 4.3).

b. First article inspection. First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 4.4).

c. Quality conformance inspection. Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification (see 4.5).

d. Quality conformance verification inspection. Quality conformance verification inspection consists of examinations and tests performed on individual products or lots to verify conformance of the products or lots with the requirements set forth in this specification (see 4.6).

4.3 Qualification inspection. The qualification inspection of the rigid seat survival kits shall consist of examinations and tests for all of the requirements of this specification and the applicable detail specification.

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4.3.1 Qualification samples. Unless otherwise specified in the applicable detail specification, the qualification samples shall consist of five complete rigid seat survival kits with oxygen, two container shells without oxygen of each manufacturer's part number submitted for qualification testing, and three elastomer test slabs, 6 by 6 by 0.075 inches, composed of identical ozone-resistant composition and cure that was used in manufacturing the elastomer components. The qualification samples shall be forwarded to the test facility set forth in the letter of authorization to submit samples (see 6.4). The samples shall be plainly identified by securely attached durable tags marked with the following information:

Samples for qualification inspection
Rigid Seat Survival Kit Type
Manufacturer's designation and number
Name of manufacturer
Submitted by (name) (date) for qualification in accordance with the requirements of MIL-S-81018 under Authorization (reference authorizing letter and number) (see 6.4).

4.3.1.1 Unless otherwise directed, a single complete kit, complying in all respects to the specification requirements herein, may be submitted prior to the submittal of the other samples required. This kit shall be forwarded to the Commander, Naval Air Development Center, Warminster, PA 18974, Code 603 and will be subjected to the dynamic crash load test, 4.8.7. Upon successful completion of this test, the manufacturer will be notified and the additional samples shall be submitted as specified in 4.3.1 for the completion of the required qualification tests. There shall be no changes in the design construction or manufacturing processes between this kit and subsequent samples.

4.3.2 Retention. The retention of qualification shall consist of periodic verification to determine compliance of the qualified product with the requirements of this specification. The time and method of periodic verification shall be specified by the activity responsible for the Qualified Products List and shall be included in the Notice of Qualification letter.

4.4 First article. The first article inspection of the rigid seat survival kits shall consist of those examinations and tests deemed necessary by the inspection facility to determine compliance with the requirements of this specification and applicable detail specification.

4.4.1 First article samples. Unless otherwise specified, as soon as practicable after award of the contract or order, the manufacturer shall submit five complete kits, one oxygen cylinder, one reducer, one charging valve and two each test slab samples, one from the contoured lid

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and one from the kit container bottom to determine the resin content. The samples shall be representative of the construction, workmanship, components and materials to be used during production. Two complete kit samples shall be forwarded to the Commander, Naval Air Development Center, Warminster, PA 18974, Code 603 and will be subjected to the dynamic ejection load test, 4.8.6, and the dynamic crash load test, 4.8.7. Three remaining samples shall be furnished to the Government as directed by the contracting officer (see 6.2d). When a manufacturer is in continuous production of these rigid seat survival kits from contract to contract, "submission of further first article" samples may be waived at the discretion of the Government procuring activity (see 6.2c). Approval of the first article samples or the waiving of first article inspection does not preclude the requirements of submitting to the quality conformance inspection, 4.5. First article samples shall be furnished to the Government as directed by the contracting officer (see 6.2). The samples shall be plainly identified by securely attached durable tags marked with the following information:

> Samples submitted by (name) (date) for first article inspection in accordance with the requirements of MIL-S-81018C(AS) under Contract No.

4.4.2 Upon completion of the first article inspection program, recommendations and comments pertinent for use in monitoring production shall be forwarded by the Government activity responsible for conducting the inspection program (see 6.2d) to the contracting officer. One approved rigid seat survival kit shall be returned to the manufacturer for use in monitoring production. The other rigid seat survival kits shall be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract.

4.5 Quality conformance inspection. The sampling and inspection levels shall conform to MIL-STD-105. Quality conformance inspection shall consist of the examinations and tests specified in table I and 4.6.

4.5.1 Sampling.

4.5.1.1 Inspection lot.

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4.5.1.1.1 Rigid seat survival kit. An inspection lot size shall be expressed in units of one rigid seat survival kit, complete with all components. An inspection lot shall consist of rigid seat survival kits, complete with all components made under identically the same conditions and from the same materials and components. The sample unit shall be one rigid seat survival kit complete with all components.

4.5.1.1.2 Packaging. An inspection lot shall be expressed in units of one fully prepared shipping container, containing rigid seat survival kits. An inspection lot shall consist of shipping containers fully prepared for delivery from essentially the same materials and components. The sample unit shall be one shipping container, containing rigid seat survival kits, fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for tests and examinations of the rigid seat survival kit. The sampling size, acceptance criteria, tests, and examinations required for the rigid seat survival kits shall be as specified in table 1.

4.6 Quality conformance verification inspection. Upon completion of the tests and examinations specified in 4.5.1.2, a random sample shall be selected from each lot in accordance with MIL-STD-105, Inspection Level S-3 by the cognizant Government quality assurance representative/specialist (see 6.2j). The sample size shall be based only on the applicable size code letter corresponding to Inspection Level S-3. Each kit selected as a sample unit shall be identified by its assigned serial number (see 6.2f) and shall be forwarded to the laboratory specified in the procurement document (see 6.2e). Each rigid seat survival kit selected as a sample shall be subjected to the following examinations and tests:

Visual examination	4.8.1.1
Release handle test	4.8.2
Kit actuation test (bench)	4.8.4.1
High pressure leakage	4.8.8.1
Reduced pressure leakage	4.8.8.2
Manual actuation	4.8.9.1
Automatic actuation	4.8.9.2
Pressure and flow	4.8.9.4
Electric continuity	4.8.10

The serial numbers of the units in the lot, represented by the sample units, shall be furnished to the laboratory. A rejected lot shall not be resubmitted without the approval of the contracting officer (see 6.6). Upon completion of the tests and examinations, the laboratory shall return the seat kit samples to the manufacturer at the manufacturer's expense. The Government activity responsible for conducting the inspection program

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(see 6.2e) shall report the results of tests and examinations to the designated inspection acceptance office specified in the procurement document. Final acceptance of the lot from which the samples were selected shall be based upon successful completion of the inspection program by the cognizant Government quality assurance representative/specialist applying the applicable acceptance criteria specified in table I.

4.7 Test conditions.

4.7.1 Atmospheric conditions. Unless otherwise specified, tests shall be conducted at local ambient temperatures and barometric pressure, and corrections shall be made to provide agreement with the temperature and pressure calibration of the instruments. Test data provided by any instrument not calibrated to Standard Temperature and Pressure (STP) conditions shall be corrected to determine STP requirements. The STP conditions are 29.92 inches of mercury and 70°F (21.1°C).

4.7.2 Test equipment. The facilities of the Naval Air Development Center, Code 603 and the Naval Aerospace Recovery Facility, El Centro CA 92243 shall be utilized in performing some of the required tests in order to preclude the purchasing of costly duplicate test equipment. An ejection seat will be required to perform some of the examinations and tests.

4.8 Inspection methods.

4.8.1 Visual examination.

4.8.1.1 Rigid seat survival kit. Every rigid seat survival kit shall be examined visually and dimensionally checked to determine conformance to all the requirements of this specification and applicable drawings. The classification of defects, table II, shall be used to classify the defects found.

4.8.1.2 Packaging. Each of the fully prepared shipping containers, containing rigid seat survival kits selected as a sample unit from the lot, shall be examined to determine that the packaging, packing, and marking conform to this specification. The classification of defects, table III, shall be used to enumerate the defects found.

4.8.2 Release handle test. A downward force of 50 pounds ± 2 percent shall be applied to the release handle of the rigid seat survival kit. The downward force shall not unlock the container. The handle shall unlock the container when a pull of 20 ± 10 pounds is applied upward on the handle. The location for the point of application of the load shall be in accordance with 3.6.2.4. After unlocking, it shall require a maximum pull of 30 pounds to detach the handle from the container.

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4.8.3 Drop test. A test dummy shall be fitted with an integrated torso harness and with a rigid seat survival kit. The weight of the kit shall be as specified in the applicable detail specification. The kit shall be attached to the torso harness by the RSSK to the torso harness restraint subassembly. This configuration shall be subjected to simulated parachute opening shock by being dropped to the end of a suspension assembly and/or drop cable of such height and elasticity as to produce 25 plus 5, minus 0, g force as recorded in the dummy. After sustaining this test, the kit shall show no evidence of distortion or damage to the restraint webbing, related hardware or installed equipment and shall satisfy the functional requirements of 4.8.2, 4.8.4 and 4.8.8.1.

4.8.4 Kit actuation test. A human or test dummy, fitted with the proper integrated torso harness and a survival kit, shall be suspended at a point approximately 35 feet above the floor to simulate actual parachute descent. Actuation of the release handle with a single sustained pull shall result in the following performance functions:

a. Release of the container locking mechanism and separation of container bottom and contoured container lid. For convenience during this test, it shall not be necessary to inflate the life raft as the container bottom falls away from the lid. The dropline shall sustain the loads imposed by the container bottom as it reaches its full extent without failure or damage. Failure of the above operations shall be cause for rejection of the rigid seat survival kit assembly.

4.8.4.1 Bench actuation test. A bench check shall be performed to verify the kits ability to unlock and for the container halves to separate properly when the release handle is actuated.

4.8.5 Static loads tests.

4.8.5.1 Condition I. A load of 2000 pounds ± 5 percent shall be applied as follows: The seat kit shall be restrained in a static load fixture simulating 50 percentile buttocks. The 2000 pound load shall then be applied to the seat kit lid through the torso harness attachments. This load should be distributed equally to both sides of the lid for a period of 5 seconds. After maintaining this load, the seat kit shall be tested and conform to the requirements of 4.8.2, 4.8.4.1 and 4.8.8.1. Any damage to the restraint system or slippage of adjustments shall be cause for rejection. Any damage to the stitching/webbing shall be evaluated by the activity responsible for coordinating tests applying the criteria set forth in table II.

4.8.5.1.1 Condition II. The container assembly shall be suspended between the torso harness attachment hardware and the ejection seat hardware of the seat to RSSK and torso harness restraint subassembly. A collective load of 6500 pounds ± 5 percent shall be applied at the rate of 4 ± 1/2

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inches per minute and maintained for five seconds. Any damage to the restraint system or slippage of adjustment shall be cause for rejection. Any damage to the stitching/webbing shall be evaluated by the activity responsible for coordinating tests applying the criteria set forth in table II.

4.8.5.2 Compression tests. Compression tests shall be conducted for vertical loading as well as fore and aft loading on the container. In each case, a proof load and/or an ultimate load shall be applied. The proof load shall be accomplished prior to the ultimate load test. The cushion shall be removed from the container for these tests. The applied loads shall be within ± 2 percent of the specified values.

4.8.5.2.1 Vertical. The compression test sample shall be placed horizontally on a compression tester so that the load shall be applied vertically into the lid. A rubber block, having a minumum thickness of one inch, a Shore "A" durometer hardness of approximately 60 and conforming to the contour of the entire container lid, shall be used. A compression load of 7250 pounds proof shall be applied vertically and downward for a period of 2 seconds, compressing the survival unit between the base of the test stand and the force apparatus. After the 7250 pound load has been relieved, the kit shall be examined. Any cracking or delamination shall be cause for rejection. Slight distortion, warping, etc., shall be acceptable. The kit shall then satisfy the function requirements for 4.8.2 and 4.8.4.1.

4.8.5.2.2 Compression strength (fore and aft). The compression test sample shall be placed on the compression tester so that the bottom of the kit is vertical. Rubber blocks which conform to the entire front and rear faces of the container shall be used to distribute the load. The rubber blocks shall be a minimum thickness of one inch and have a Shore "A" durometer hardness of approximately 60. Compression loads of 2000 pounds proof and 3000 pounds ultimate shall be applied vertically and downward for a period of 2 seconds, compressing the container between the base of the test stand and the force apparatus. After the 2000 pound proof load is relieved, the kit shall show no evidence of distortion or damage to the container or installed equipment and shall satisfy the functional requirements of 4.8.2 and 4.8.4.1. After the 3000 pound ultimate load has been applied and removed, the kit shall be examined. Any cracking or delamination shall be cause for rejection. Slight distortion or warping shall be acceptable.

4.8.6 Dynamic ejection load tests. These ejections shall be conducted under simulated emergency conditions on an ejection tower. An anthropomorphic dummy weighing 250 pounds shall be placed on the kit and ejected at a maximum acceleration of 20 ± 1 g and rate of change acceleration not to exceed 250 g per second. Ejection phenomenon shall not exceed 200 milliseconds. The Naval Air Development Center, Code 603, has an ejection tower which is capable of producing this ejection load type phenomenon. There shall be no evidence of any cracking or distortion of the kit. Following this test, the kit shall be subjected to and meet the requirements of the following tests: 4.8.2, 4.8.4.1 and 4.8.8.1.

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4.8.7 Dymanic crash load tests. The container shall be capable of withstanding crash type loads supporting a 250 pound anthropomorphic dummy in the ejection seat. The minimum acceleration load and rate of change of acceleration shall be as described in figure 2. This test shall be conducted with the load applied through the kit lid at any and all angles between "Z" axis (downward) forward and aft to the "X" axis and within 20° of the "X-Z" plane as depicted in figure 2. Attitude shall be as described in the applicable detail specification. Following this test, the kit shall be subjected to and meet the requirements of the following tests: 4.8.2, 4.8.4.1 and 4.8.8.1.

4.8.8 Oxygen system leakage.

4.8.8.1 High pressure leakage. The emergency oxygen system, charged to at least 1800 psig, shall be stored for 24 hours with the oxygen valve in closed position. The oxygen system shall be checked for leakage by use of a detecting solution. There shall be no leakage as evidenced by the L-2 gage and leak detecting solution before and after the time period.

4.8.8.2 Reduced pressure leakage. Apply 100 psig pressure first to the AN833-4D fitting and check the entire low pressure system, for leakage with the drawing CL223A3 fitting capped. Second, apply 110 psig pressure to the drawing CL223A3 (RSSK-8 type application or equivalent) fitting and check for leakage past the check valve, from the AN833-4D fitting. The leakage, including relief valve leakage, shall not exceed 0.02 lpm.

4.8.8.3 Oxygen low pressure leakage. With the oxygen system charged to 250 psig maximum and with the oxygen reducer switch in the off position, there shall be no leakage when determined by testing with leak detector fluid.

4.8.8.4 Oxygen relief pressure. The relief valve shall be tested to insure valve opening at 120 to 140 psig and the reset at 110 psig minimum. The relief valve shall not leak before or after test.

4.8.9 Emergency oxygen supply.

4.8.9.1 Manual actuation. The on/off actuator of the pressure reducer shall be actuated when a pull force of 20 ± 10 pounds is applied upward on the manual actuating "O" ring of the emergency bailout oxygen system, with 0 psig and 90 psig applied to the reducer outlet (90 psig simulates aircraft's supply).

4.8.9.2 Automatic actuation. The on/off actuator of the pressure reducer shall be actuated when a pull force of 20 ± 10 pounds is applied to the pull cable of the automatic actuation connection, with 0 psig and 90

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psig applied to the reducer outlet (90 psig simulates aircraft's supply). Disconnection must occur after actuation of the reducer.

4.8.9.3 Endurance. The emergency oxygen supply actuation tests shall be repeated 25 times in the manual condition and 25 times in the automatic condition to insure correct operation of the pressure reducer. Do not apply any pressure against the reducer outlet during the endurance test. At the completion of the 50 endurance cycles, the emergency oxygen supply shall be subjected to and pass the requirements of 4.8.9.1 and 4.8.9.2.

4.8.9.4 Pressure and flow. The seat kit oxygen hose shall be attached to the seat kit and then connect the proper mating oxygen hose to the kit hose. Using inlet pressures of 250 and 1800 psig in the oxygen cylinder, set flows of 0 and 90 lpm from the reducer and determine the outlet pressure at the end of the mating hose. The pressure shall be a minimum of 40 psig.

4.8.10 Electrical continuity. With the oxygen hose installed in the seat kit, check the electrical continuity of the integrated electrical wire. The integrated electical wire shall have electrical continuity.

4.8.11 Dielectric strength. Electrical insulation shall be capable of withstanding repeated applications of test voltage of 500 volts (RMS), at commercial frequency, between isolated terminals and between the case and all isolated terminals connected together, for at least one minute. There shall be no evidence of arcing. Failure shall be evidenced by arcing or leaking current changes, large rapid increase (insulation breakdown), slow gradual increase, and/or fluctuation.

4.8.12 Environmental tests. A rigid seat survival kit shall be subjected to each of the following environmental tests:

4.8.12.1 Vibration test. The test sample shall be mounted on the vibration stand in a position similar to the normal aircraft mounting. Vibration tests shall be conducted and continually monitored, if applicable, under both resonant and cycling conditions in accordance with procedure XII of MIL-E-5272, except that the vibration range curve shall follow a constant double amplitude of 0.036 inches from 5 to 75 cps and a constant amplitude of plus or minus 10 g from 75 cps to 500 cps. The oxygen cylinder shall be pressurized to 1800 psig and a pressure of 90 psig shall be applied to the AN833-4D fitting to simulate the aircraft's supply pressure. At the completion of each axis test, the kit shall be subjected to and meet the requirements of 4.8.2 and 4.8.4.1. In addition, the unit shall be inspected thoroughly and shall not have sustained any damage or malfunction resulting from the vibration tests. After the vibration test, the kit shall pass the requirements of 4.8.8.1. The amplitude or acceleration for the frequency cycling shall be within plus or minus 10 percent of the specified values. The vibration test shall be performed at room temperature.

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4.8.12.2 Extreme temperature.

4.8.12.2.1 High temperature operation. The kit test sample with the oxygen cylinder charged to at least 1800 psig shall be subjected to 160° ± 3.6°F for a period of 3 hours. Immediately upon removal from the temperature chamber, the kit shall be subjected to and meet the requirements of 4.8.2 and 4.8.4.1. The oxygen system shall also be checked for leakage by use of a leak detecting solution. The test sample shall then be allowed to return to room temperature, at which time, the gage pressure shall be checked. There shall be no evidence of pressure loss. The kit shall then be subjected to and meet the requirements of 4.8.8.2.

4.8.12.2.2 Low temperature operation. The kit test sample with the oxygen cylinder charged to at least 1800 psig shall be subjected to -65° ± 3.6°F for a period of 3 hours. Immediately upon removal from the temperature chamber, the kit shall be subjected to and meet the requirements of 4.8.2 and 4.8.4.1. The oxygen system shall also be checked for leakage by use of a leak detecting solution. The test sample shall then be allowed to return to room temperature, at which time, the gage pressure shall be checked. There shall be no evidence of pressure loss. The kit shall then be subjected to and meet the requirements of 4.8.8.2.

4.8.12.3 Humidity test. The complete rigid seat survival kit shall be subjected to humidity in accordance with procedure I of MIL-E-5272, for not less than 120 hours and shall then be subjected to and meet the requirements of 4.8.2, 4.8.4.1 and 4.8.8 through 4.8.8.4.

4.8.12.4 Salt spray tests. The complete rigid seat survival kit shall be subjected to salt spray in accordance with procedure I of MIL-E-5272, for not less than 120 hours and shall then be subjected to and meet the requirements of 4.8.2, 4.8.4.1 and 4.8.8 through 4.8.8.4.

4.8.12.5 Sand and dust test. The complete rigid seat survival kit shall be subjected to sand and dust in accordance with procedure I of MIL-E-5272 and shall then be subjected to and meet the requirements of 4.8.2, 4.8.4.1 and 4.8.8 through 4.8.8.4.

4.8.13 Resin content. Two test slab samples, one from the kit contoured lid and the other from the kit container bottom, shall be used to determine the resin content of the rigid seat survival kit container. The location, where the samples shall be taken, shall be at the discretion of the testing activity and shall be consistent with the size of the specimens required by FED-STD-406. The test slabs shall be tested in accordance with FED-STD-406, method 7061 and shall exhibit the resin content specified in 3.4.3.2.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified by the procuring activity (see 6.2).

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5.1.1 Level A or C. The rigid seat survival kits shall be packaged in accordance with MIL-STD-794, as specified by the procuring acitivity (see 6.2). For Level A packaging, the kits shall be packaged in accordance with MIL-P-116, method III, utilizing containers conforming to PPP-B-636, type I or II, class 2, grade 3.

5.2 Packing. The rigid seat survival kit shall be packed Level A, B, or C, in accordance with MIL-STD-794, as specified by the procuring activity (see 6.2). As far as practicable, exterior containers shall be of uniform size and shape and be of minimum cube and tare consistent with the protection required.

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.1 Precautionary marking. The following precautionary marking shall appear on each unit package:

"CAUTION: DO NOT ALLOW OIL OF ANY KIND, CONTACT PRESERVATIVE OR HYDROCARBON MATERIALS TO BE USED IN OR ABOUT THE RIGID SEAT SURVIVAL KIT."

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. The rigid seat survival kits are intended for use with the applicable ejection seat. The rigid seat survival kit is used in the ejection seat to limit the dynamic center of gravity shift of the aircrewman during ejection, thereby reducing the tendency to tumble during the propulsion phase of the ejection by limiting the center of gravity travel of the seat-man combination within the resultant thrust vector of the propulsion system.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.

b. Quantity required.

c. Whether first article inspection is waived (see 4.4.1).

d. Name and address of the first article laboratory (see 4.4.1), including the name and address of the Government activity responsible for conducting the first article inspection program (see 4.4.1).

e. Name and address of the quality conformance verification inspection laboratory (see 4.6),

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including the name and address of the Government activity responsible for conducting the quality conformance verification inspection program (see 4.6).

f. Issuance of block of serial numbers to cover quantity of kits to be purchased (see 4.6.).

g. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

h. Whether any special markings are required.

i. Items of data required (see 6.3).

j. Selection of samples to be performed by the cognizant Government quality assurance representative/ specialist (see 4.6).

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9 (n) of Defense Acquisition Regulations (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
4.4.1	First Article Inspection Reports	DI-T-5329 - Inspection Test Reports

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they might be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command, Department of the Navy, Washington, DC 20360; however, authorization for qualification of products shall be obtained from the Commanding Officer, Naval Air Development Center, Warminster, Pa 18974, (Code 6031). Prior to submission of the

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samples for qualification inspection, the manufacturer shall submit a request to the Naval Air Development Center (Code 6031) indicating a date on which the samples can be forwarded and also request an authorization number to accompany the samples.

6.5 First article. When a first article is required, it shall be tested under the appropriate provision of 7-104.55 of the Defense Acquisition Regulation. The first article should be a preproduction sample. The contracting officer should include specified instructions for examinations, tests and approval of the first article.

6.6 Laboratory information. The successful bidder will be furnished with the name of the quality conformance verification inspection laboratory. The costs of the tests and examinations on samples initially submitted from a lot, as required by 4.6, shall be borne by the Government. Samples from a rejected lot shall not be resubmitted for tests and examinations, as required by 4.6, without the approval of the contracting officer. The costs of the tests and examinations on samples resubmitted from a reworked lot or from a new lot which is necessitated by the rejection of a previous lot shall be borne by the manufacturer.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing Activity:
Navy - AS
(Project No. 1660-N406)

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TABLE I. Sample size, acceptance criteria, tests, and examinations of the rigid seat survival kit.

Inspection	Method paragraph	Sample size	Acceptance criteria
Visual examination	4.8.1.1	Every rigid seat survival kit for critical defects. Inspection Level II for minor defects	Reject all units with any critical defect and an acceptable quality level of 2.5 defects per hundred units for minor defects
Release handle test	4.8.2	Inspection Level S-4 1/	Acceptance number zero rejection number 1
Kit actuation test (bench)	4.8.4.1	Inspection Level S-4 1/	Acceptance number zero rejection number 1
High pressure leakage	4.8.8.1	Inspection Level S-4 1/	Acceptance number zero rejection number 1
Reduced pressure leakage	4.8.8.2	Inspection Level S-4 1/	Acceptance number zero rejection number 1
Manual actuation	4.8.9.1	Inspection Level S-4 1/	Acceptance number zero rejection number 1
Automatic actuation	4.8.9.2	Inspection Level S-4 1/	Acceptance number zero rejection number 1

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TABLE I. Sample size, acceptance criteria, tests, and examinations of the rigid seat survival kit. - continued.

Inspection	Method paragraph	Sample size	Acceptance criteria
Pressure and flow	4.8.9.4	Inspection Level S-4 1/	Acceptance number zero rejection number 1
Electrical continuity	4.8.10	Inspection Level S-4 1/	Acceptance number zero rejection number 1
Packaging	4.8.1.2	Inspection Level S-2	Total acceptable quality level of 2.5 defects per hundred units

1/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.

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TABLE II. Classification of defects for visual examination of the rigid seat survival kit.

CRITICAL	MINOR
101. Any material imperfection, foreign matter embedded, sliver, burr, or sharp edge inside or outside the kit.	201. Any marking missing, insufficient, incorrect, illigible, or not permanent.
102. Any surface unclean, rough, misaligned or containing any crack, nick or other flaw.	202. Any color not as specified or does not approximately match specified color.
103. Any component missing, malformed, fractured, or otherwise damaged.	203. Color of any component not as specified or does not approximately match color of container.
104. Any frayed or cut webbing or strap.	
105. Any component loose or otherwise not securely retained.	
106. Incorrect assembling or improper position of any component.	
107. Any functioning part that works with difficulty.	
108. Any faulty workmanship or other irregularities.	
109. Any oil, contact preservative, or hydrocarbon material on any of the emergency oxygen system components or hoses.	
110. Any measurement not within specified dimension and tolerance.	
111. Any spot or stain. 1/	
112. Any corrosion, scale, pit or dent.	
113. Any fibrous component that contains any tear, cut, hole, needle chew, burn, patch, mend or abraded area.	

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TABLE II. Classification of defects for visual examination of the rigid seat survival kit. - continued

CRITICAL	MINOR
114. Any component not as specified or any defect of component or defect of assembly, not herein classified. 1/	
115. Any component, component part or required operation omitted or any operation improperly performed, not herein classified. 1/	

1/ The defect shall be classified and scored as a critical defect when it seriously affects the serviceability or appearance, otherwise it shall be classified and scored as a minor defect.

TABLE III. Classification of defects for packaging.

ITEM	DEFECTS
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence or method of application; markings not the same on the interior and exterior containers.
Workmanship	Inadequate application of the components, such as incomplete closure of the unit package, intermediate package, container flaps, or loose strappings, etc.; bulging or distortion of the containers.
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirement.

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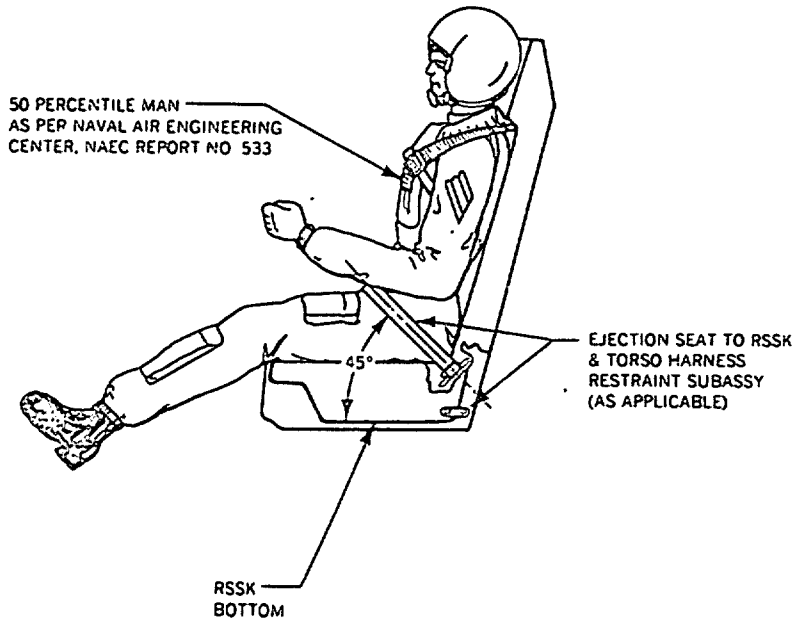
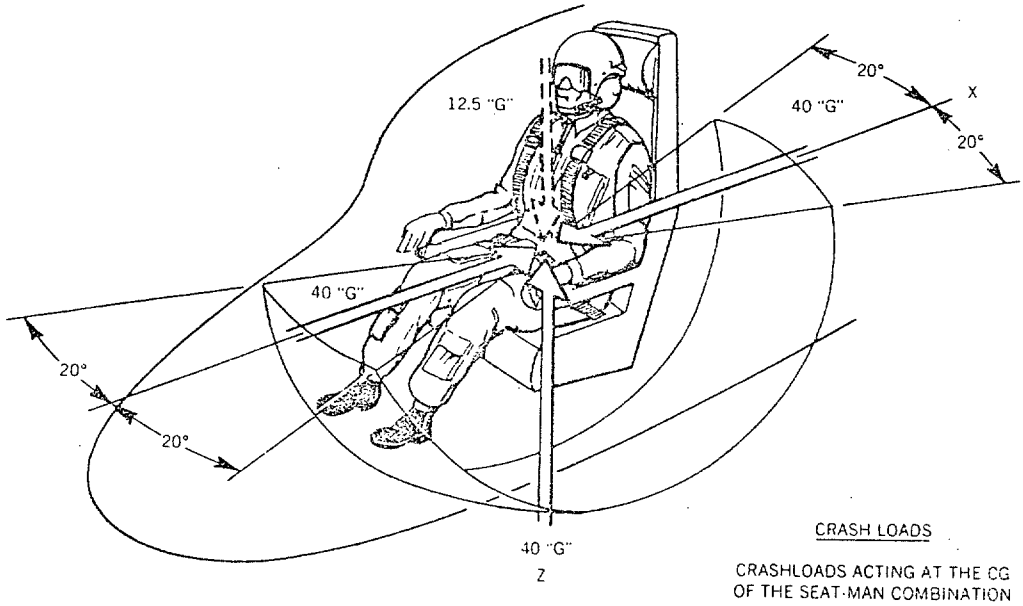


FIGURE 1. Optimum design arrangement of the ejection seat to RSSK & torso harness restraint subassy.

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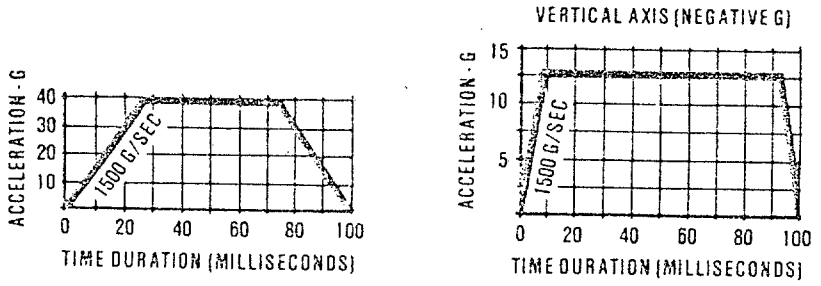


FIGURE 2. Dynamic crash loads

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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